Guangzhou Food Guide Automated Content Management System

Problem Analysis & Theme

The project addresses the challenge of managing and automatically generating restaurant review reports for a Guangzhou food guide. The core requirements include:

1. **Data Management**: Store and manage restaurant information including ratings, prices, locations, and recommended dishes
2. **Automated Report Generation**: Create dynamic reports based on predefined templates using database queries
3. **Template System**: Support flexible template-based content generation in multiple formats (text, markdown, HTML)
4. **Advanced Data Analysis**: Perform complex queries using aggregate functions and subqueries to extract meaningful insights

The system needed to handle 299+ restaurant records with various attributes such as star ratings, price ranges, cuisine categories, business districts, and multi-dimensional ratings (taste, environment, service).

Solution Approach

Architecture Overview

The system follows a three-tier architecture:

1. **Data Layer**: SQLite database
2. **Backend Layer**: Flask RESTful API providing data access and business logic
3. **Frontend Layer**: React SPA

Key Technologies

- **Database**: SQLite 3 with foreign key constraints and indexes
- **Backend**: Python 3.13, Flask, SQLAlchemy-style DAO pattern
- **Frontend**: React 18, TypeScript, Vite
- **Data Processing**: Pandas for CSV import/export, custom template rendering engine

Methodology and Implementation Techniques

1. Database Design Pattern

Entity Relationship Diagram:

```
restaurants (1) ──┐
                  │
                  ├──< (N) reports (N) >── (1) templates
                  │
meta (standalone configuration table)
```

Key Design Principles:

- **Normalization**: Separate concerns between data (restaurants), templates (report structures), and outputs (generated reports)
- **Soft Deletion**: Use `is_active` flag instead of physical deletion to maintain data integrity
- **Version Control**: Template versioning for audit trail
- **Metadata Storage**: Flexible key-value storage for system configuration

2. Advanced SQL Query Implementation

The system implements complex queries using:

Common Table Expressions (CTEs):

```
WITH filtered AS (
    SELECT star_level, taste_score, environment_score, service_score
    FROM restaurants WHERE is_active = 1
),
level_counts AS (
    SELECT star_level, COUNT(*) AS cnt FROM filtered GROUP BY star_level
),
level_avg AS (
    SELECT star_level, AVG(taste_score) AS avg_taste FROM filtered GROUP BY star_level
)
SELECT * FROM level_counts, level_avg...
```

Aggregate Functions:

- `COUNT(*)` for statistical counting
- `AVG()` for average calculations (taste, environment, service scores)
- `MAX()`, `MIN()` for range analysis
- `SUM()` for cumulative metrics

Subqueries:

- Correlated subqueries for filtering based on aggregated values
- Scalar subqueries for computed columns
- EXISTS subqueries for conditional record retrieval

3. Template Engine Architecture

Template Storage:

- Templates stored in database with JSON metadata for placeholders
- Support for dynamic content generation with `{placeholder}` syntax
- Multi-format output (text, markdown, HTML)

Rendering Process:

1. Load template from database
2. Execute query service based on template_id
3. Build context dictionary from query results
4. Replace placeholders using Python's `string.Formatter`
5. Generate outputs in all supported formats
6. Store generated report (optional)

Template Categories:

1. **Star Rating Analysis**: Aggregate statistics with subqueries
2. **Best Value Recommendations**: Sorting with calculated metrics
3. **Seasonal Recommendations**: Filtered queries (winter warmers)
4. **Hidden Gems**: Complex filtering with multiple conditions
5. **District Overviews**: Group-by aggregations
6. **Delivery Recommendations**: Composite scoring

4. API Design Patterns

RESTful Endpoints:

- `GET /api/restaurants/`: List with pagination, filtering, sorting
- `POST /api/restaurants/`: Batch insert/update
- `GET /api/restaurants/<id>`: Single record retrieval
- `PUT /api/restaurants/<id>`: Update operation
- `DELETE /api/restaurants/<id>`: Soft delete
- `GET /api/templates/`: Template management
- `POST /api/reports/generate`: Dynamic report generation
- `GET /api/statistics/*`: Aggregated statistics

Data Access Object (DAO) Pattern:

- Separation of concerns: DAO handles database operations
- Service layer manages business logic
- API routes handle HTTP request/response

5. Frontend State Management

Client-Side Optimizations:

- Server-side pagination (reduces data transfer)
- Request caching and cancellation
- Debounced search inputs
- Optimistic UI updates
- Loading states with skeleton screens

Component Architecture:

- Reusable components (StarRating, LoadingSpinner, Toast)
- Custom hooks for data fetching (useRestaurants)
- Error boundaries for graceful error handling
- Dark mode support with CSS variables

Database Structure and Queries

Table Structure

restaurants (Primary Entity):

```
CREATE TABLE restaurants (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    shop_id TEXT NOT NULL UNIQUE,
    name TEXT NOT NULL,
    reviews INTEGER DEFAULT 0,
    avg_price REAL,
    category TEXT NOT NULL,
    business_district TEXT,
    address TEXT,
    recommended_dish_1 TEXT,
    recommended_dish_2 TEXT,
    recommended_dish_3 TEXT,
    taste_score REAL,
    environment_score REAL,
    service_score REAL,
    star_level TEXT,
    url TEXT,
    is_active INTEGER DEFAULT 1,
    created_at TEXT DEFAULT CURRENT_TIMESTAMP,
    updated_at TEXT DEFAULT CURRENT_TIMESTAMP
);
```

templates (Report Templates):

```
CREATE TABLE templates (
    template_id INTEGER PRIMARY KEY,
    title TEXT NOT NULL,
    keywords TEXT,
    default_format TEXT NOT NULL CHECK (default_format IN ('text', 'markdown', 'html')),
    content TEXT NOT NULL,
    placeholders_json TEXT NOT NULL,
    sample_output TEXT,
    version INTEGER DEFAULT 1,
    updated_at TEXT DEFAULT CURRENT_TIMESTAMP
);
```

reports (Generated Reports):

```
CREATE TABLE reports (
    report_id INTEGER PRIMARY KEY AUTOINCREMENT,
    template_id INTEGER NOT NULL REFERENCES templates(template_id) ON DELETE CASCADE,
    title TEXT NOT NULL,
    params_json TEXT NOT NULL,
    content_text TEXT,
    content_markdown TEXT,
    content_html TEXT,
    created_at TEXT DEFAULT CURRENT_TIMESTAMP
);
```

Indexes:

- Performance optimization on frequently queried columns:
  - `category`, `business_district`, `star_level`, `avg_price`
  - Foreign key index on `reports.template_id`

Triggers:

- Automatic `updated_at` timestamp on record modification

Queries

Static statistical query

- Distribution of Cuisines (Top 10)
- Business District Distribution (Top 10)
- Price distribution
- Star distribution
- Top 10 Popular Restaurants

Dynamic report query

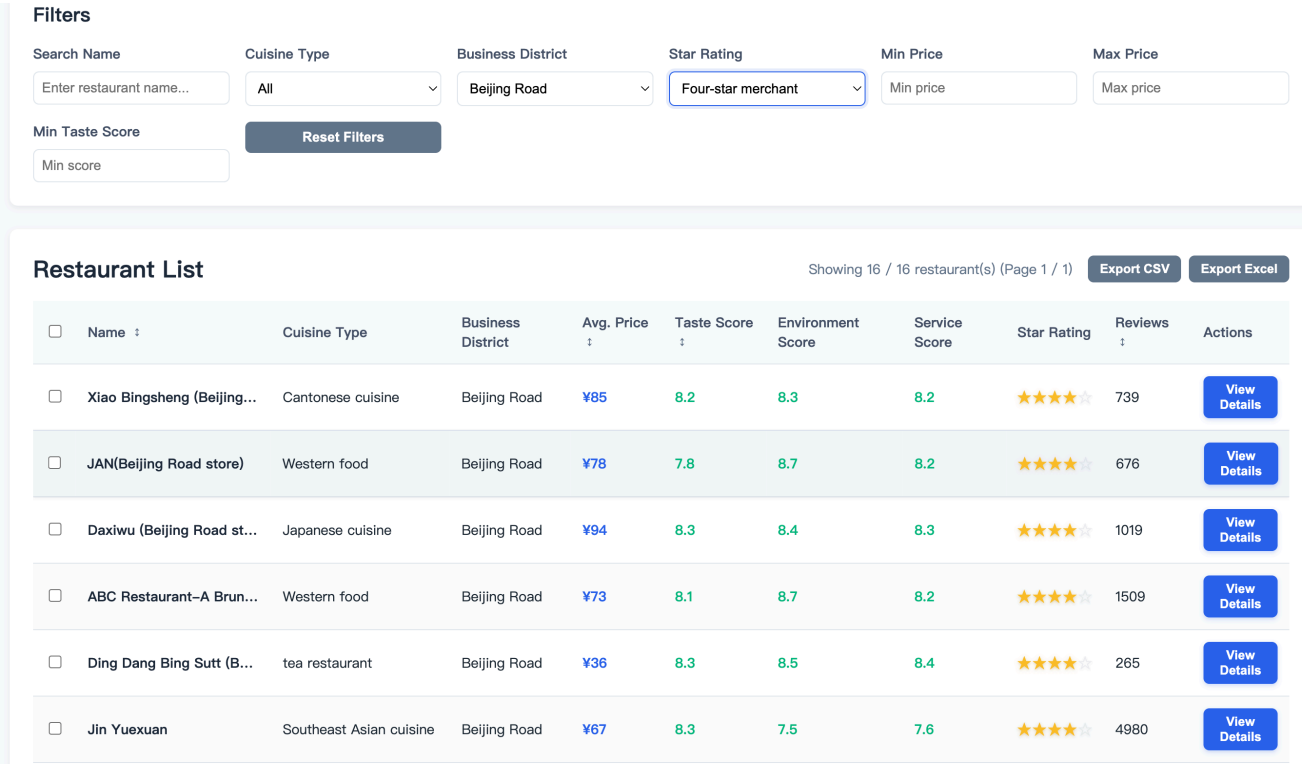
- Star rating distribution and scoring performance (including aggregation and subqueries)

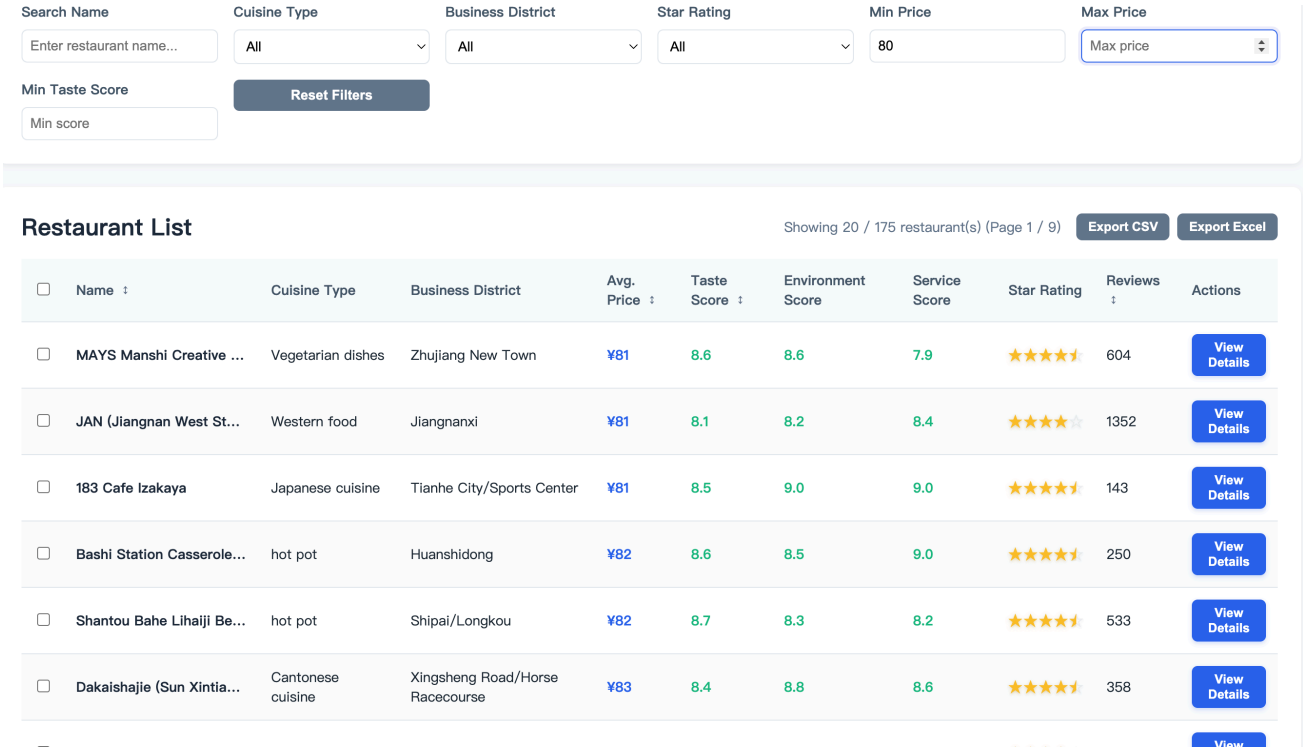
- Calculate the cost performance based on the taste score and price
- Get recommendations for hot pot and stews in the designated business district
- Select treasure restaurants with fewer reviews but higher ratings than the regional average
- Regional overview, showcasing mainstream cuisines and representative restaurants
- Choose takeout friendly restaurants with a comprehensive score higher than the threshold

Function

CRUD

- Main page
- Aggregated search





- Show restaurant detail

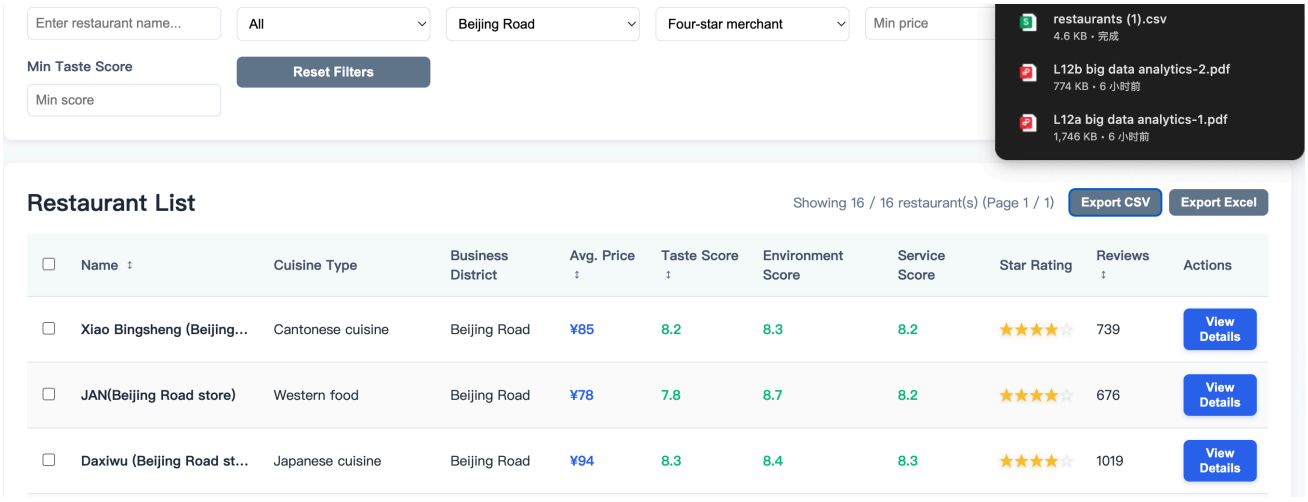
Xiao Bingsheng (Beijing Road store)

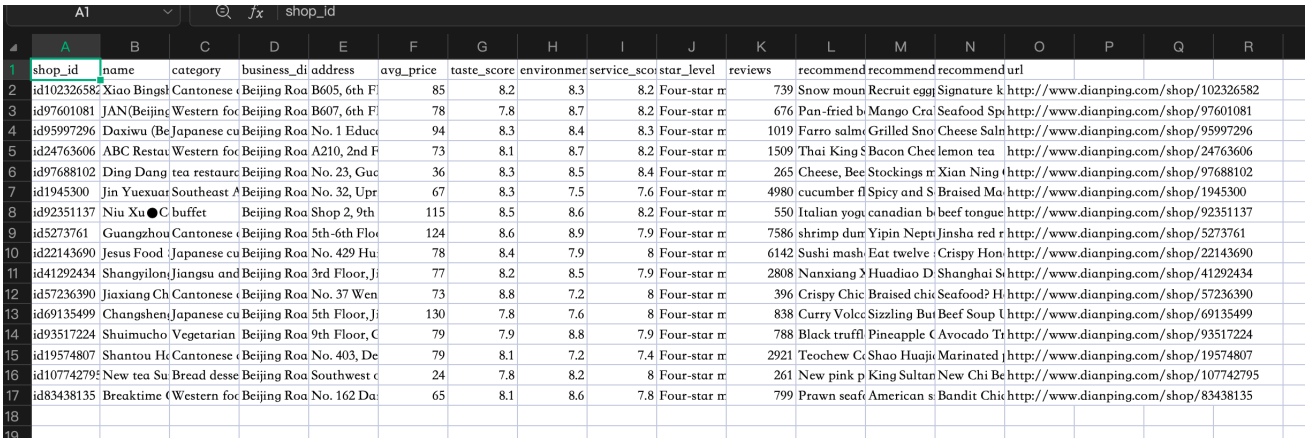
Shop ID:	id102326582
Cuisine Type:	Cantonese cuisine
Business District:	Beijing Road
Address:	B605, 6th Floor, Yangzhonghui, Guangdonghai, No. 168 Beijing Road
Average Price:	¥85
Taste Score:	8.2
Environment Score:	8.3
Service Score:	8.2
Star Rating:	★★★★☆ Four-star merchant
Reviews:	739

- Edit restaurant



- Delete restaurant
- Export page information





shop_id	name	category	business_di	address	avg_price	taste_score	environmer	service_sco	star_level	reviews	recommend	recommend	recommend	url
id102326582	Xiao Bingsl	Cantonese (	Beijing Roa	B605, 6th F	85	8.2	8.3	8.2	Four-star n	739	Snow moun	Recruit egg	Signature k	http://www.dianping.com/shop/102326582
id97601081	JAN(Beijing	Western foo	Beijing Roa	B607, 6th F	78	7.8	8.7	8.2	Four-star n	676	Pan-fried b	Mango Cra	Seafood Sp	http://www.dianping.com/shop/97601081
id95997296	Daxiwu (Be	Japanese cu	Beijing Roa	No. 1 Educ	94	8.3	8.4	8.3	Four-star n	1019	Farro salm	Grilled Sno	Cheese Saln	http://www.dianping.com/shop/95997296
id24763606	ABC Restau	Western foo	Beijing Roa	A210, 2nd F	73	8.1	8.7	8.2	Four-star n	1509	Thai King S	Bacon Chee	lemon tea	http://www.dianping.com/shop/24763606
id97688102	Ding Dang	tea restaurc	Beijing Roa	No. 23, Guc	36	8.3	8.5	8.4	Four-star n	265	Cheese, Bee	Stockings n	Xian Ning (	http://www.dianping.com/shop/97688102
id1945300	Jin Yuexuar	Southeast A	Beijing Roa	No. 32, Upr	67	8.3	7.5	7.6	Four-star n	4980	cucumber fl	Spicy and S	Braised Ma	http://www.dianping.com/shop/1945300
id92351137	Niu Xu●C	buffet	Beijing Roa	Shop 2, 9th	115	8.5	8.6	8.2	Four-star n	550	Italian yogu	canadian b	beef tongue	http://www.dianping.com/shop/92351137
id5273761	Guangzhou	Cantonese (	Beijing Roa	5th-6th Floo	124	8.6	8.9	7.9	Four-star n	7586	shrimp dun	Yipin Neptı	Jinsha red r	http://www.dianping.com/shop/5273761
id22143690	Jesus Food	Japanese cu	Beijing Roa	No. 429 Hu	78	8.4	7.9	8	Four-star n	6142	Sushi mash	Eat twelve	Crispy Hon	http://www.dianping.com/shop/22143690
id41292434	Shangyilon	Jiangsu and	Beijing Roa	3rd Floor, Ji	77	8.2	8.5	7.9	Four-star n	2808	Nanxiang X	Huadiao D	Shanghai S	http://www.dianping.com/shop/41292434
id57236390	Jiaxiang Ch	Cantonese (	Beijing Roa	No. 37 Wen	73	8.8	7.2	8	Four-star n	396	Crispy Chic	Braised chic	Seafood? H	http://www.dianping.com/shop/57236390
id69135499	Changshen	Japanese cu	Beijing Roa	5th Floor, Ji	130	7.8	7.6	8	Four-star n	838	Curry Volc	Sizzling But	Beef Soup U	http://www.dianping.com/shop/69135499
id93517224	Shuimucho	Vegetarian	Beijing Roa	9th Floor, C	79	7.9	8.8	7.9	Four-star n	788	Black truffl	Pineapple (	Avocado Tr	http://www.dianping.com/shop/93517224
id19574807	Shantou Hc	Cantonese (	Beijing Roa	No. 403, De	79	8.1	7.2	7.4	Four-star n	2921	Teochew Cc	Shao Huaji	Marinated ;	http://www.dianping.com/shop/19574807
id107742795	New tea Su	Bread desse	Beijing Roa	Southwest (	24	7.8	8.2	8	Four-star n	261	New pink p	King Sultan	New Chi Be	http://www.dianping.com/shop/107742795
id83438135	Breaktime (	Western foo	Beijing Roa	No. 162 Da	65	8.1	8.6	7.8	Four-star n	799	Prawn seafc	American s	Bandit Chic	http://www.dianping.com/shop/83438135

Show static statistical query

- Overall Statistics

Overall Statistics

Total Restaurants	Cuisine Types	Business Districts	Average Price	Avg. Taste Score
299	27	50	¥119	8.5

Avg. Environment Score	Avg. Service Score	Total Reviews
8.5	8.4	490449

- Cuisine Distribution (Top 10)

Cuisine Distribution (Top 10)

Cuisine Type	Count	Avg. Price	Avg. Taste Score	Avg. Environment Score	Avg. Service Score
Western food	50	¥155	8.4	8.8	8.7
Cantonese cuisine	42	¥101	8.5	8.2	8.2
Japanese cuisine	41	¥155	8.6	8.5	8.7
hot pot	30	¥99	8.6	8.2	8.3
Bread dessert	18	¥114	8.5	8.5	8.6
buffet	16	¥252	8.3	8.8	8.6
barbecue	14	¥74	8.4	8.3	8.2
cafe	12	¥68	8.2	8.7	8.6
Korean food	9	¥82	8.5	8.4	8.4
tea restaurant	9	¥61	8.8	8.6	8.4

- Business District Distribution (Top 10)

Business District Distribution (Top 10)

Business District	Count	Avg. Price	Avg. Taste Score
Zhujiang New Town	49	¥144	8.3
Tianhe City/Sports Center	38	¥118	8.5
Beijing Road	26	¥84	8.3
Tianhebei	20	¥145	8.5
Jiangnanxi	18	¥80	8.4
Airport Road	9	¥89	8.7
Jiangnan Avenue	8	¥80	8.4
Nancun	8	¥114	8.3
Xingsheng Road/Horse Racecourse	8	¥115	8.6
industrial avenue	8	¥139	8.4

- Price Distribution

Price Distribution

Price Range	Count	Percentage
Below ¥50	30	10.0%
¥50–100	143	47.8%
¥100–150	61	20.4%
¥150–200	24	8.0%
¥200–300	22	7.4%
Above ¥300	19	6.4%

- Star Rating Distribution

Star Rating Distribution

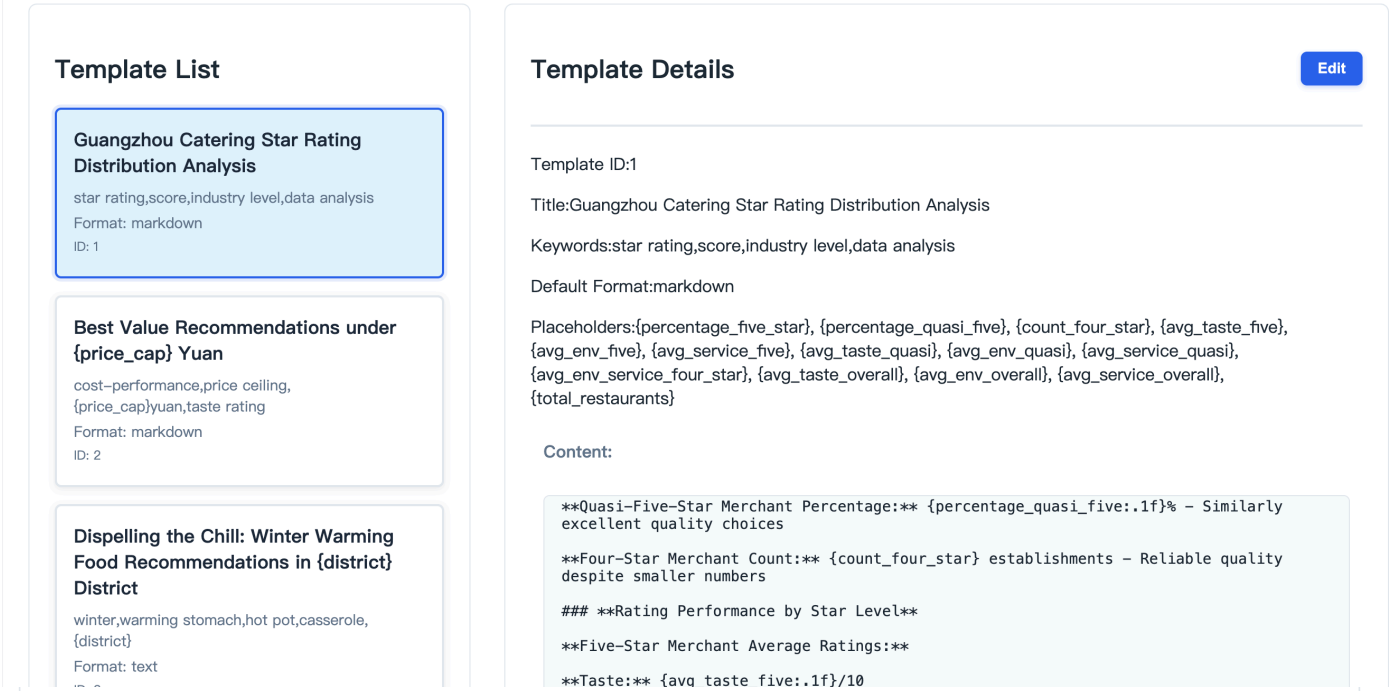
Star Rating	Count	Avg. Taste Score	Avg. Environment Score	Avg. Service Score
★★★★½ Quasi five-star merchant	132	8.6	8.6	8.6
★★★★☆ Four-star merchant	131	8.2	8.3	8.1
★★★★★ Five-star merchant	33	9.0	9.0	9.1
★★★½☆ Quasi four-star merchant	3	7.5	7.8	7.4

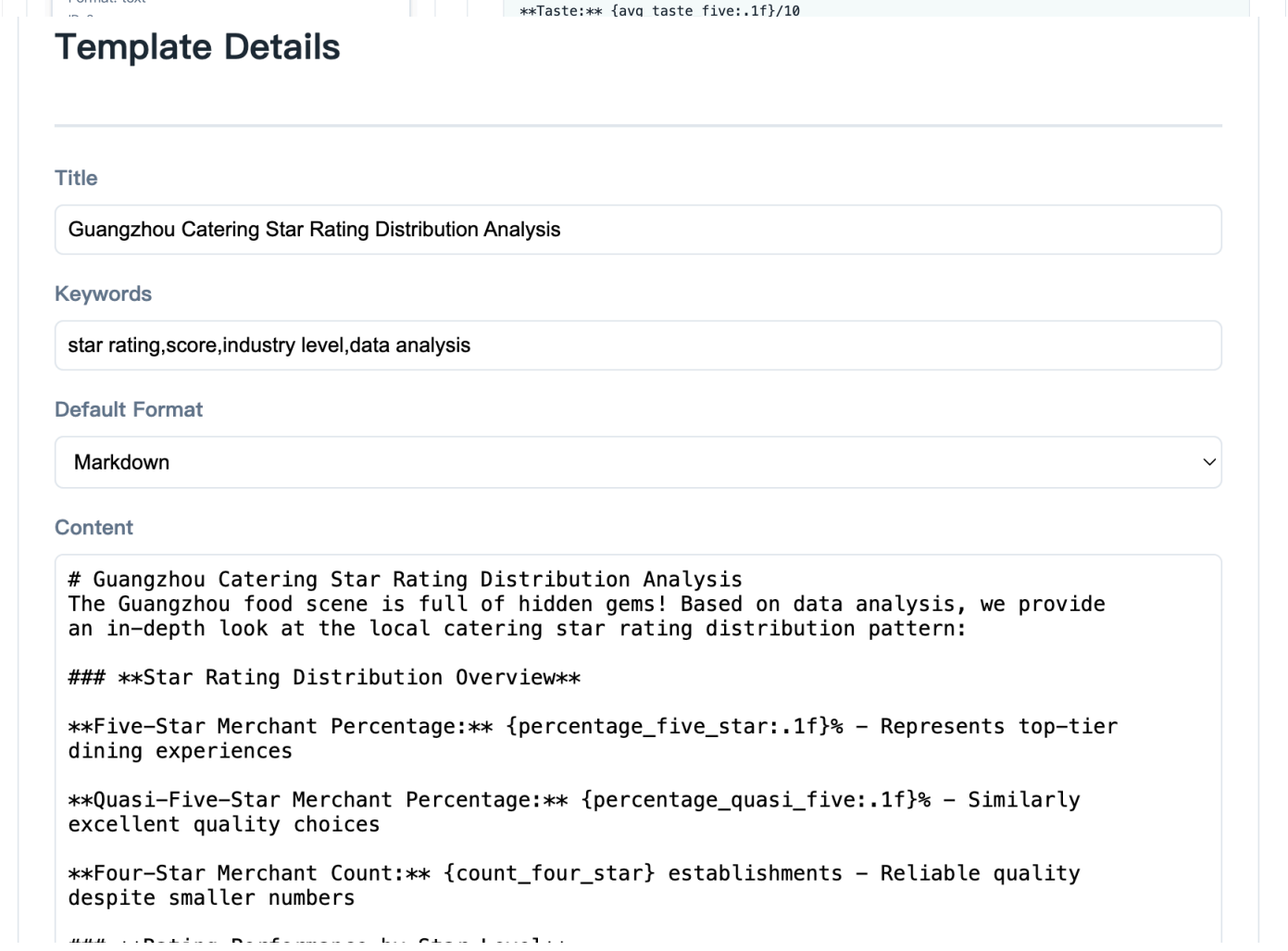
- Popular Restaurants (Top 10)

Popular Restaurants (Top 10)

Restaurant Name	Cuisine Type	Business District	Price	Taste Score	Environment Score	Service Score	Star Rating	Reviews
Da Lu Ye (Grandview Mall Branch)	Cantonese cuisine	Tianhe City/Sports Center	¥54	9.3	9.1	9.1	★★★★★	151
Kushipachi·Robatayaki	Japanese cuisine	Xingsheng Road/Horse Racecourse	¥178	9.3	9.0	9.0	★★★★★	70
Eternal health steam pot	seafood	Shiqiao	¥111	9.2	8.2	9.1	★★★★★	648
Artisan baking	Bread dessert	Kecun/Chigang	¥156	9.2	9.1	9.2	★★★★★	729
Blossom Bakery Studio	Bread dessert	Tianhe City/Sports Center	¥187	9.2	9.1	9.2	★★★★★	278
Extremely hot Taiwanese exquisite charcoal barbecue	Japanese cuisine	Tianhe City/Sports Center	¥211	9.1	8.5	9.2	★★★★★	3551
Manhuotang·Healthy Organic Western Restaurant (Gaozhi Branch)	Western food	Xingsheng Road/Horse Racecourse	¥148	9.1	9.0	9.0	★★★★★	1174
Dian Du De (Jun He Lou store)	tea restaurant	Shiqiao	¥79	9.1	9.0	8.8	★★★★★	1393
ZENHO Tenwa Kappo Cuisine	Japanese cuisine	Zhujiang New Town	¥528	9.1	9.3	9.3	★★★★★	508
Ruyue Exquisite Cuisine●Internet celebrity check-in hotspot	Japanese cuisine	Tianhebei	¥101	9.1	9.2	9.1	★★★★★	169

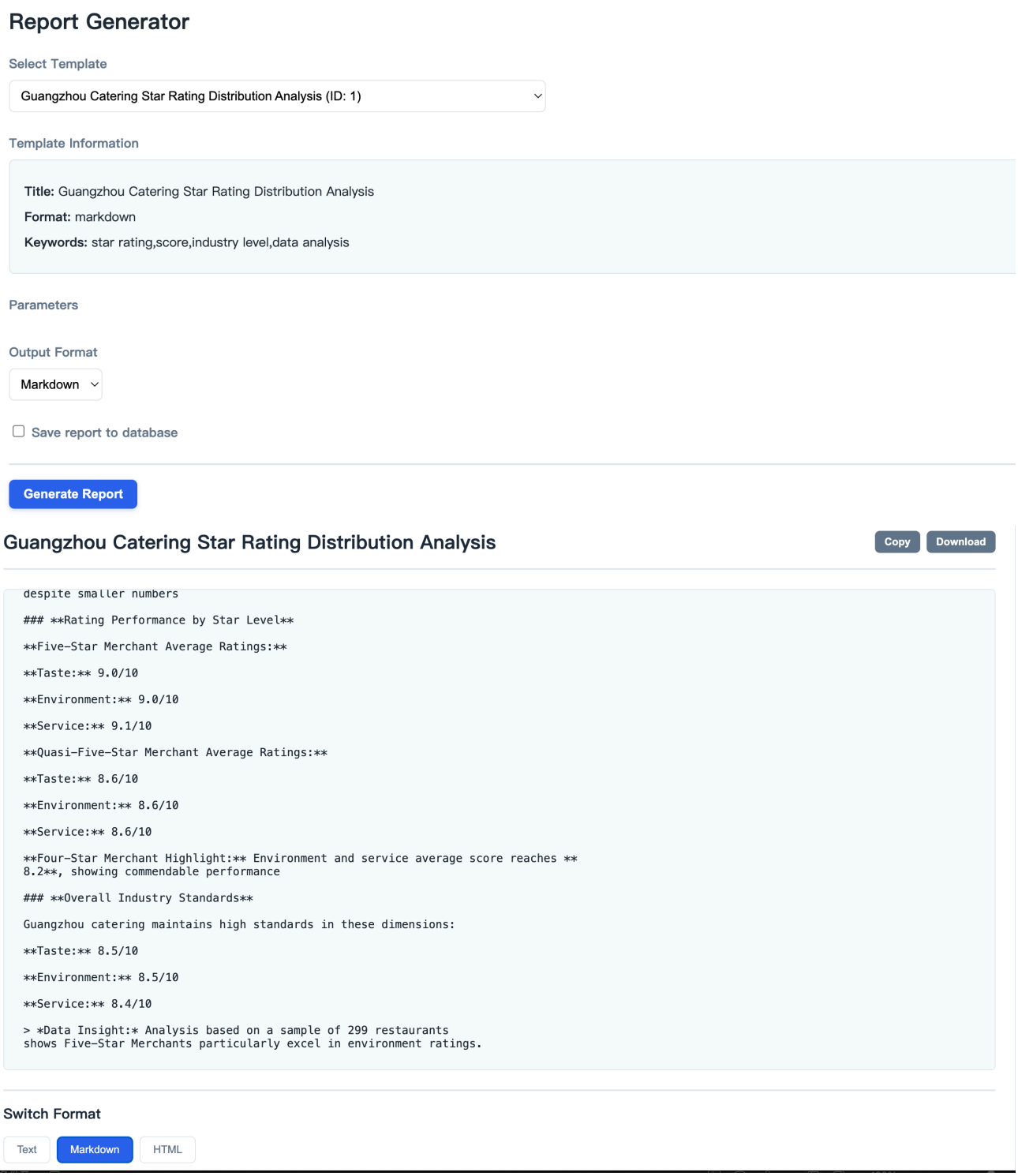
Temperate management



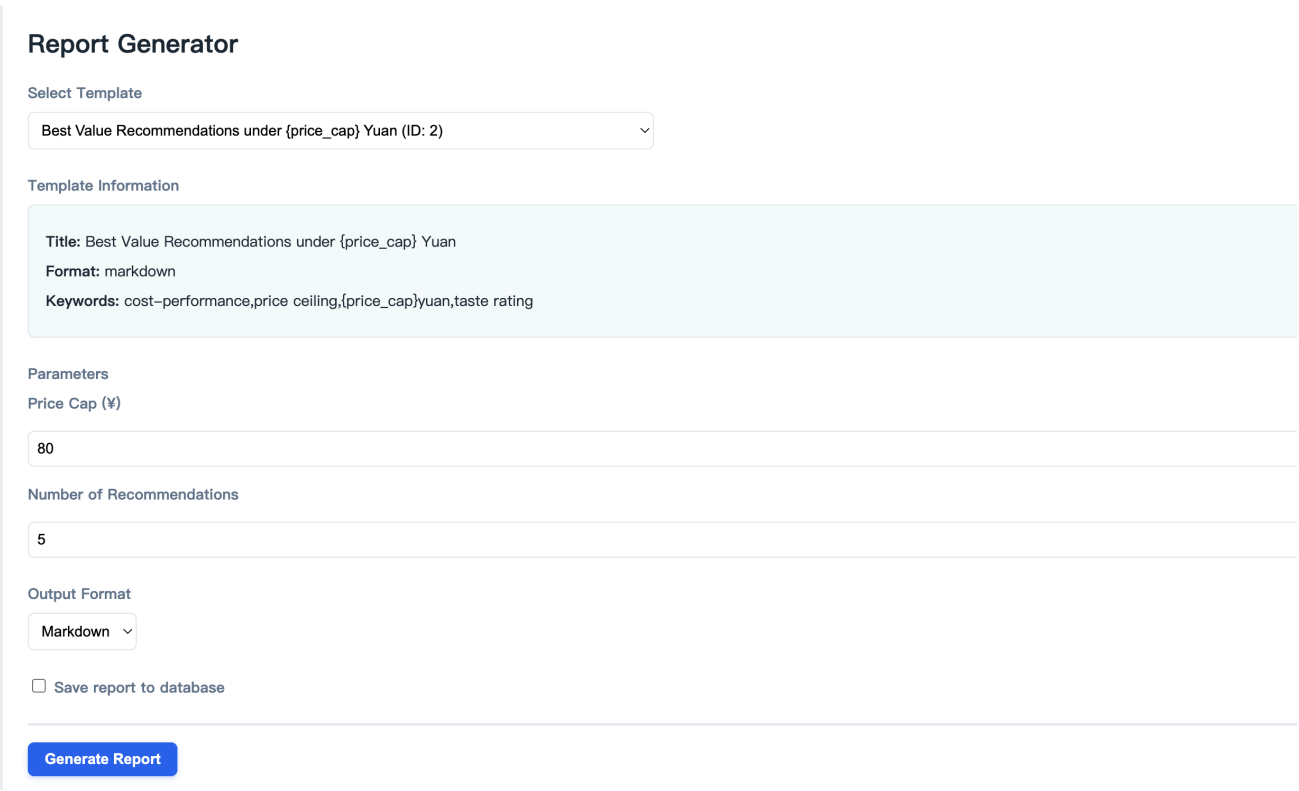


Report generator

- Star rating distribution and scoring performance



- Best value recommendations under X yuan



Best Value Recommendations under 80.0 Yuan

Copy Download

```
# Best Value Recommendations under 80.0 Yuan
Based on your budget, we've selected the 5 restaurants with the best value for
money under 80.0 yuan! (Value determined by taste rating to price ratio)

**Taning Tart Lemon·Craft Lemon Tea (Huacheng Huinan District Store)** (Value Index: 0.67)

**Taste Rating:** 8.1

**Avg Price:** ¥12.0

**Recommended Dishes:** `Hand-beaten lemon green (Yipin green tea)`, `Hand-made Lemon Red (Keemen Black Tea)`

**Banyan Tree (Baiyun Flagship Store)** (Value Index: 0.49)

**Taste Rating:** 8.3

**Avg Price:** ¥17.0

**Recommended Dishes:** `Buddha jumps over the wall`, `Fried Chencun Noodles`

**Jincha Research Laboratory** (Value Index: 0.48)

**Taste Rating:** 9.1

**Avg Price:** ¥19.0

**Recommended Dishes:** `Cheese fresh berries`, `Fresh cup of peach`

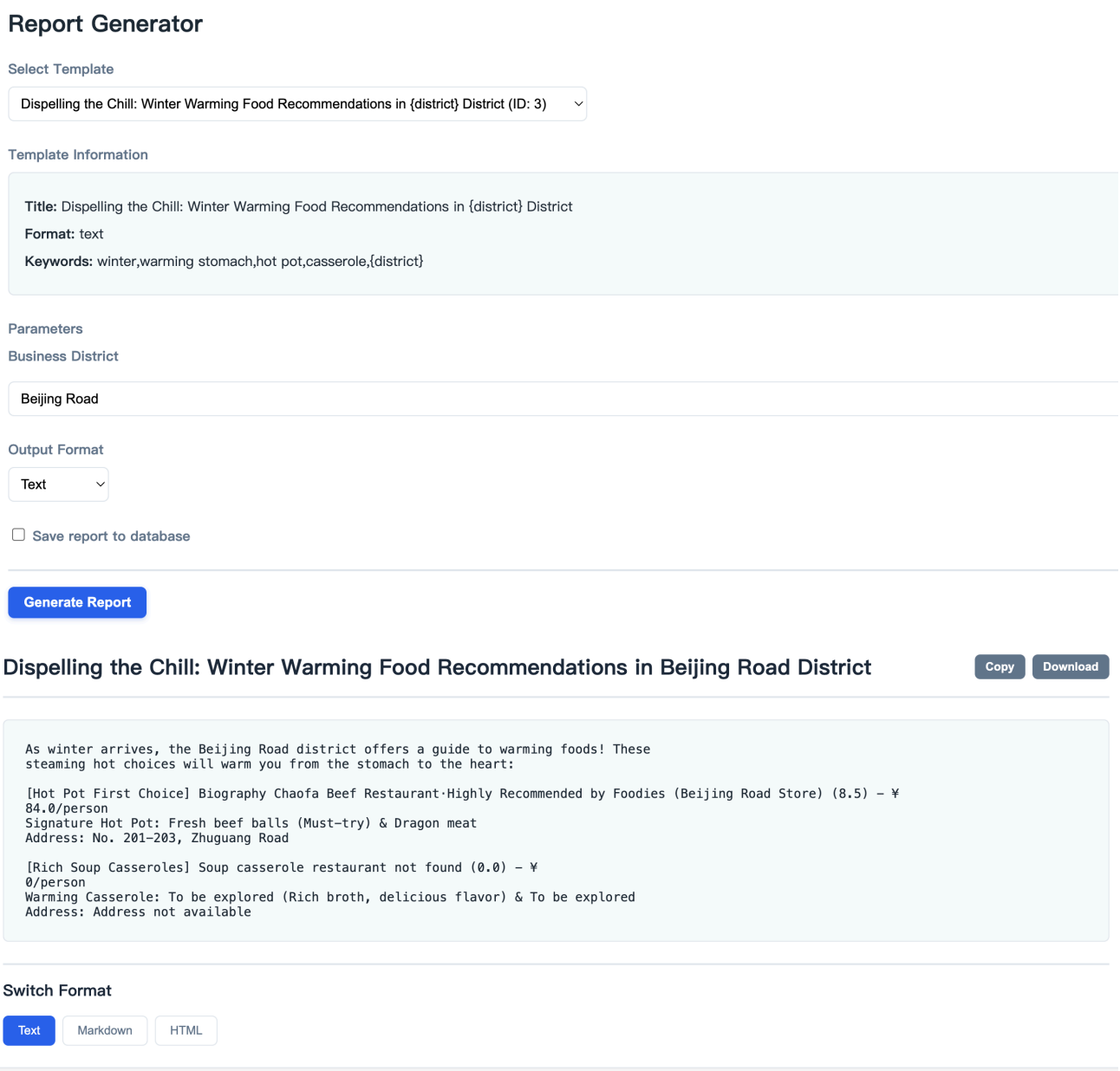
**Zhao Ji Heritage Milk Dessert Shop (Jiangyan Store)** (Value Index: 0.44)

**Taste Rating:** 8.3

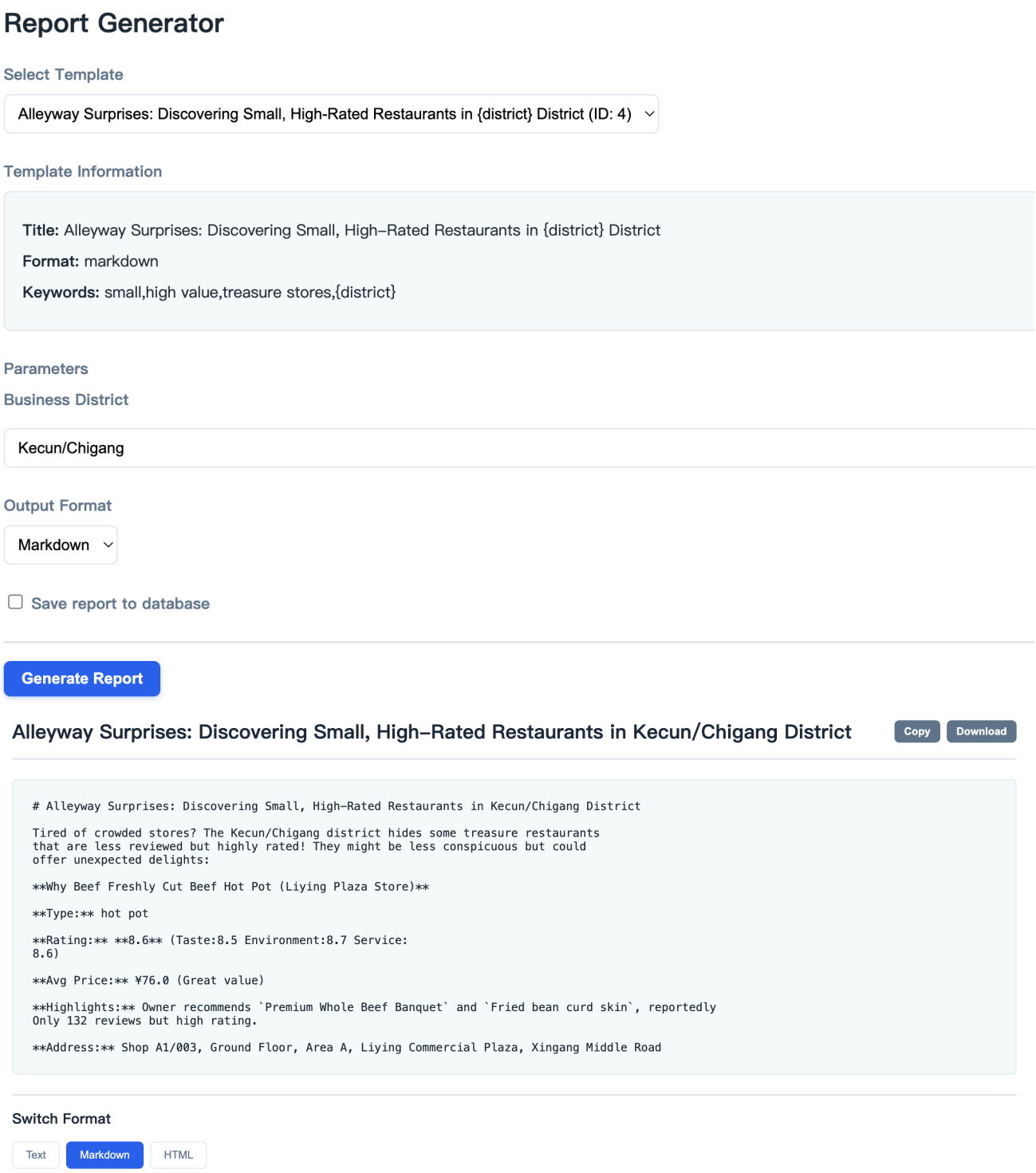
**Avg Price:** ¥19.0

**Recommended Dishes:** `Ginger hits milk`, `milk bricks`
```

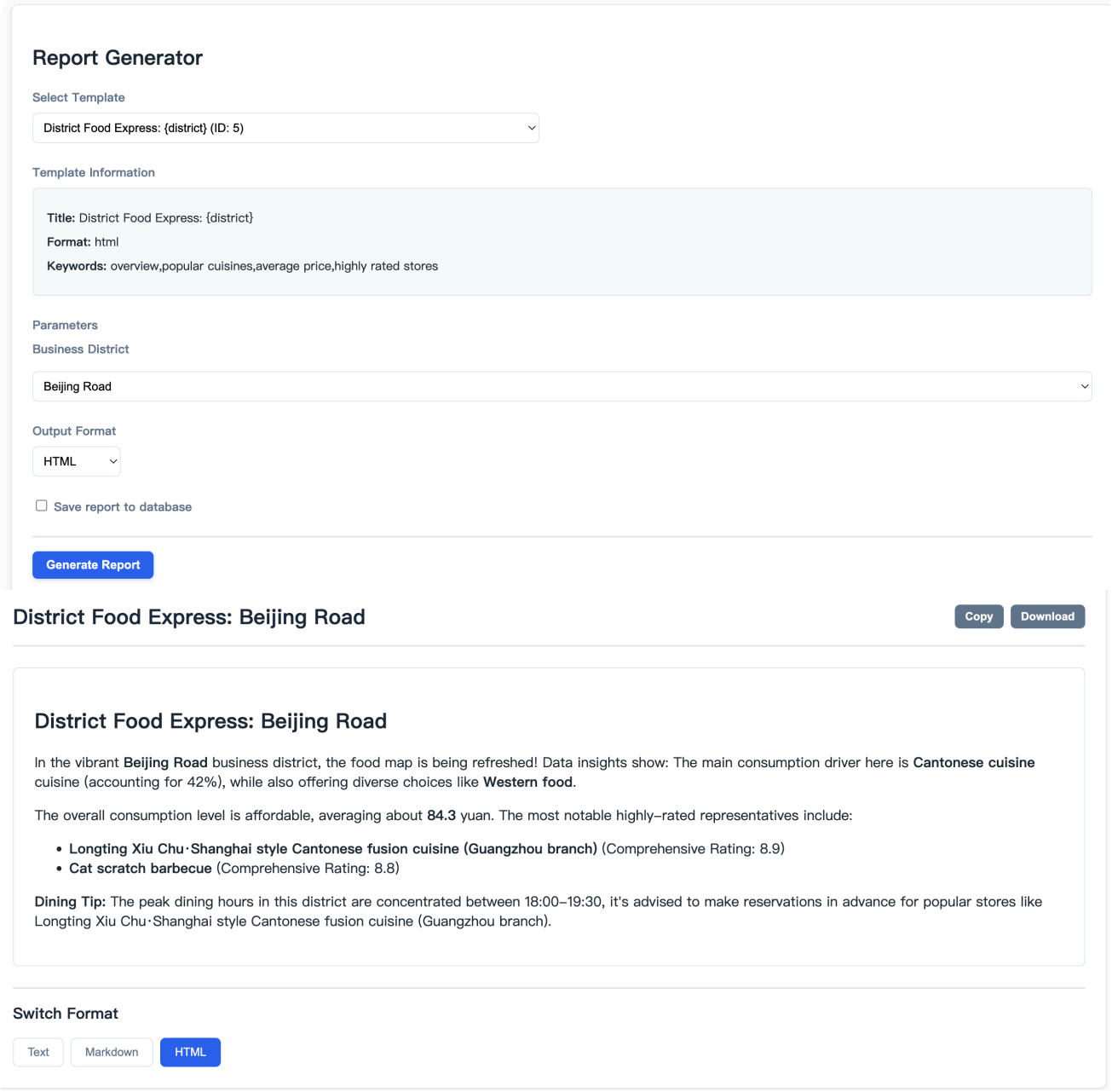
- Winter warming food recommendations



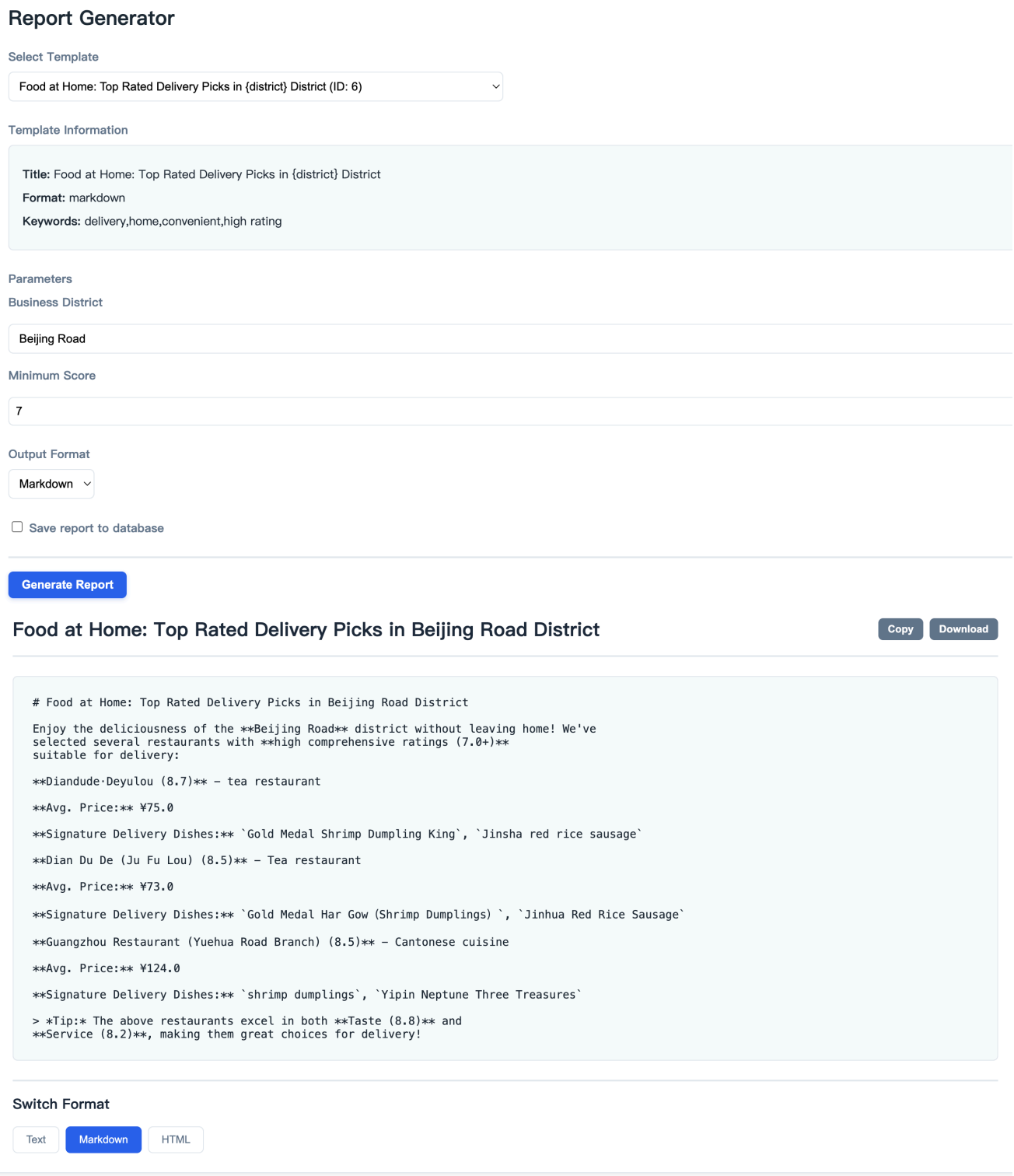
- Alleyway Surprises: Discovering Small, High-Rated Restaurants



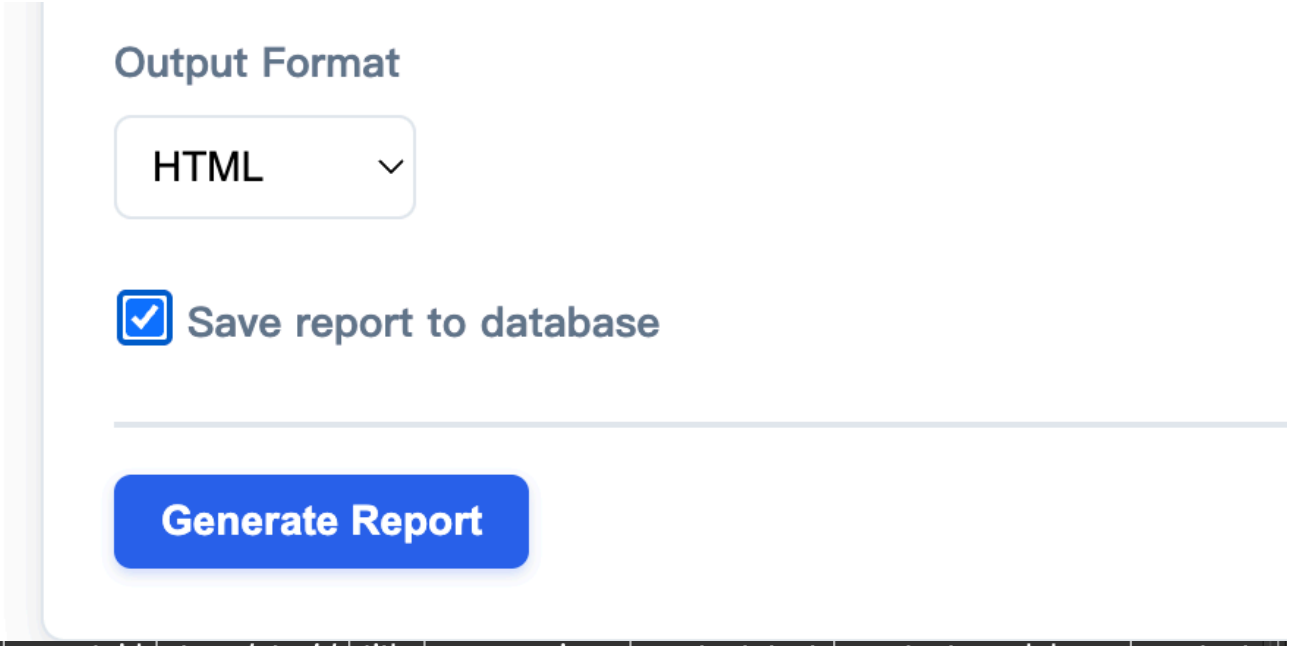
- District food express



- Food at home: Top Rated Delivery Picks



- Export report
- Save in database



report_id	template_id	title	params_json	content_text	content_markdown	content_
过滤	过滤	过滤	过滤	过滤	过滤	过滤
1	1	Gu...	{}	# Guangzhou ...	# Guangzhou Catering ...	<h1>Guang
2	6	Fo...	{"district": ...	# Food at Hom...	# Food at Home: Top ...	<h1>Food a

Results and Findings

Performance Metrics

- **Data Volume**: Successfully manages 299 active restaurant records
- **Template System**: 6 pre-defined report templates covering various analysis scenarios
- **Query Performance**: Complex aggregation queries execute in <50ms with proper indexing
- **API Response Time**: Average 30-80ms for paginated restaurant lists

Key Achievements

1. **Scalable Architecture**: Modular design allows easy extension of templates and query types
2. **Data Integrity**: Foreign key constraints and soft deletion maintain referential integrity
3. **Flexible Reporting**: Template-based system enables rapid report generation without code changes
4. **User Experience**: Responsive UI with real-time filtering, sorting, and export capabilities
5. **Code Quality**: Separation of concerns with DAO, Service, and API layers

Conclusion

The Guangzhou Food Guide CMS successfully demonstrates the implementation of a full-stack content management system with advanced database querying capabilities. The use of aggregate functions and subqueries enables complex data analysis, while the template-based reporting system provides flexibility for generating various types of food guide content. The project showcases effective separation of concerns,

modern web development practices, and demonstrates practical application of database design principles in a real-world scenario.